

## Code-Layout, Readability and Reusability

|                      |                           |
|----------------------|---------------------------|
| <b>Date</b>          | 1 <sup>st</sup> July 2026 |
| <b>Team ID</b>       |                           |
| <b>Project Name</b>  | Rising Waters             |
| <b>Maximum Marks</b> | 5 Marks                   |

### Code-Layout, Readability and Reusability: Code Layout Checklist

| S.No | Code Quality Parameter    | Description                                                                                   | Followed | Remarks                                    |
|------|---------------------------|-----------------------------------------------------------------------------------------------|----------|--------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing and indentation maintained throughout the project.                            | Yes      | Improves readability and coding standards. |
| 2    | Proper File Structure     | Files and folders are logically organized into templates, static files, models, and datasets. | Yes      | Modular project organization.              |
| 3    | Meaningful Variable Names | Variables clearly represent their purpose and functionality.                                  | Yes      | Easy to understand and maintain.           |
| 4    | Function / Method Names   | Functions use descriptive names that explain their operations.                                | Yes      | Enhances code readability.                 |
| 5    | Code Comments             | Important sections contain                                                                    | Yes      | Facilitates future maintenance.            |

|          |                   |                                                                                       |     |                                      |
|----------|-------------------|---------------------------------------------------------------------------------------|-----|--------------------------------------|
|          |                   | comments explaining implementation logic.                                             |     |                                      |
| <b>6</b> | Modular Design    | Application logic is divided into reusable functions and modules.                     | Yes | Supports scalability and code reuse. |
| <b>7</b> | No Redundant Code | Duplicate or unnecessary code has been removed.                                       | Yes | Keeps the application efficient.     |
| <b>8</b> | Error Handling    | User input validation and exception handling are implemented for critical operations. | Yes | Improves application reliability.    |

#### Reusable Components / Modules

| <b>S.No</b> | <b>Component / Module Name</b> | <b>Language / Technology</b> | <b>Where Reused</b>                            | <b>Reusability Level</b> |
|-------------|--------------------------------|------------------------------|------------------------------------------------|--------------------------|
| <b>1</b>    | Flask Routing Module           | Python, Flask                | Handles navigation between pages               | High                     |
| <b>2</b>    | Prediction Module              | Python, Scikit-learn         | Used for flood prediction requests             | High                     |
| <b>3</b>    | Data Preprocessing Module      | Python, StandardScaler       | Reused before every prediction                 | High                     |
| <b>4</b>    | Random Forest Model Loader     | Joblib                       | Loads trained model during application startup | High                     |
| <b>5</b>    | HTML Templates                 | HTML5, Bootstrap             | Reused across multiple                         | Medium                   |

|  |  |  |              |  |
|--|--|--|--------------|--|
|  |  |  | web<br>pages |  |
|--|--|--|--------------|--|

### Overall Code Quality Assessment

| Aspect                             | Rating (1–5) | Comments                                                                                        |
|------------------------------------|--------------|-------------------------------------------------------------------------------------------------|
| <b>Code Layout &amp; Structure</b> | 5            | Well-organized modular project structure.                                                       |
| <b>Readability</b>                 | 5            | Meaningful naming conventions and consistent formatting improve readability.                    |
| <b>Reusability</b>                 | 5            | Multiple reusable modules reduce duplication and simplify maintenance.                          |
| <b>Documentation / Comments</b>    | 4            | Important logic is documented with comments.                                                    |
| <b>Overall Score</b>               | 4.8 / 5      | Clean, maintainable, and reusable implementation following software engineering best practices. |